

THE FUTURE OF RETAIL: A STRATEGIC GUIDE TO DEMAND-SENSING WITH IOT & MACHINE LEARNING TECHNOLOGY

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ABSTRACT

Fashion retailers continue to rely on barcode-based systems that offer limited visibility, slow processes, and minimal insight into complex shopper behavior. This creates a critical gap in accurate demand forecasting and efficient inventory management, especially as fast-fashion cycles accelerate and SKU complexity (size, color, style) continues to grow. The purpose of this study is to address this gap by developing a demand sensing solution that uses IoT, RFID, and machine learning to generate real-time, SKU-level predictive insights for fashion retail environments. We designed an RFID-enabled demand sensing system based on a four-layer IoT architecture. Data are captured through item-level RFID tags, smart shelves, and fitting room sensors, then processed through AWS IoT Core, Kinesis, S3, DynamoDB, and Redshift. Five regression models are evaluated and combined into hybrid predictors that select the best-performing approaches for each SKU. Fuzzy logic is also used to interpret ambiguous customer interactions, specifically distinguishing between items browsed, tried on, but ultimately abandoned. The proposed architecture supports continuous visibility into product movement and enables automated checkout through multi-item RFID scanning, which reduces transaction times and improves the shopping experience. By integrating IoT data with machine learning techniques, this approach provides retailers with more accurate and context-aware demand insights, helping reduce out-of-stocks on popular sizes, minimize deadstock risks, and support a more data-driven fashion ecosystem.

Keywords: *Internet of Things, RFID Technology, Demand Sensing, Machine Learning, Smart Retail*

1. INTRODUCTION

The fashion retail sector is experiencing a rapid shift driven by digital transformation and the increasing demand for efficiency, transparency, and highly personalized services. One of the major enablers of this transformation is the Internet of Things (IoT), particularly Radio Frequency Identification (RFID) technology. By embedding RFID tags in apparel, footwear, and accessories, companies can track, monitor, and analyze individual stock keeping units (SKUs) in real time. This transforms traditional retail operations into data-driven ecosystems that not only streamline processes but also generate valuable insights for granular demand prediction. With the support of cloud platforms, these data streams can be securely stored and processed at scale, enabling more accurate and timely analytics.

Despite these advances, many fashion brands still rely heavily on barcode-based systems and manual inventory tracking. These conventional methods are time-consuming, error-prone, and limited in

providing meaningful insights into customer behavior. Most importantly, barcode scanning doesn't provide visibility into the "try-on" phase, the most crucial moment in clothing shopping. This visibility gap creates inefficiencies in inventory management, leads to size-level stockouts, and limits the ability to predict which styles will be successful or unsuccessful before the season ends.

This white paper explores how RFID and IoT technologies, complemented by cloud-enabled analytics, can address these challenges and reshape the fashion retail experience. Beyond improving operational efficiency and enhancing the customer journey, RFID-powered systems enable predictive analytics, thus helping retailers identify nuanced purchasing patterns, anticipate trend-driven demand, and make informed decisions on merchandising strategies to minimize deadstock and markdowns.

2. MARKET LANDSCAPE AND INDUSTRY OUTLOOK

The recent Grand View Research highlights the accelerating adoption of Internet of Things solutions in the fashion retail sector. As illustrated in the chart "Internet of Things in Retail Market, Size by Component, 2024-2033 (USD Billion)", the market is projected to expand from USD 66.4 billion in 2024 to USD 488.5 billion by 2033, reflecting a compound annual growth rate (CAGR) of roughly 24–25% over the decade. The growth of IoT Solutions seen across all IoT segments including hardware, platform, and services. But the platform and service show the most growth that indicates a clear industry shift from merely deploying connected devices toward data centric platforms and managed services that facilitate real-time decision-making, demand sensing, and advanced analytics.

This growth curve aligns with the trend where fashion retailers are leveraging cloud-based IoT architectures and machine learning analytics to achieve better and detailed visibility of shopper behavior and apparel inventory flows. The capability to support operational efficiency and predictive that is required for demand sensing leads to the increasing number of investments in smart shelves, RFID infrastructures, and edge computing that supports it all. Fashion retailers invested and prioritized to leverage scalable data platforms and AI driven insights over simple hardware rollouts. This reflects a maturing market where competitive advantages are created by how well the IoT provider company interprets and acts on IoT data.

Currently, North America holds the largest market share, according to Grand View Research, while Asia-Pacific is predicted to grow at the fastest rate. This trend is particularly significant for Southeast Asian nations like Indonesia, where urban fashion retail infrastructure is developing quickly. Together, these statistics demonstrate that the IoT fashion ecosystem is about to enter a period of continuous, innovation-driven growth. As a result, apparel merchants hoping to stay competitive through 2033 must make early investments in demand-sensing systems and predictive analytics.

Within this supply and demand cycle, the fashion retailer serves a dual purpose as service deliverer and information-based decision-maker. They not only keep shelves fully supplied but ensure that style availability and consumer experience always go together. Findings of two new studies show that consumer demands arise at both ends of the retailing relationship between fashion brand and end buyer.

The stores need an in-depth, real-time knowledge of consumer activity to enhance store format, product positioning, and virtue-based promotions. By integrating RFID-tracking-based tracking with machine learning-based classification, store managers can scrutinize consumer “shopping paths,” determine which products attract the greatest attention, and provide personalized offers directly to consumers’ devices. Such functions make data-based product positioning and advertising strategies possible, solving key store imperatives such as lowering stockouts, augmenting conversion of revenue, and enhancing consumer loyalty by harmonizing stock with actual activity within the store.

The role of predictive insight to align stock with real-time demand and thus reduce waste generated by unsold or obsolete stock. By mining IoT sensor data using association-rule mining, managers can identify best-selling combinations of products and predict demand with increased accuracy, backing operational efficiency and strategic merchandising planning. From the consumer perspective, the same IoT system streamlines the shopping experience by steering shoppers to the relevant products, making sure that fast-moving stock is always available, and customizing offers to align with the style preferences of the customers. Taking together, these observations demonstrate how demand-sensing technologies respond to the requirements of both store owners and eventually buyers and create an efficient and consumer-driven fashion ecosystem.

In this modern competitive landscape in IoT field for fashion retail is shaped by tech giants and specialized players. Nowadays, hardware companies add platform layers and cloud-based service. Intel is investing in edge computing where they able to provide processors that allow data to be analyzed in real time that reduce reliance on cloud latency. While NXP focuses on RFID and NFC chips that sit at the core of automated checkout systems and smart shelf applications. In the other hand, Zebra Technologies has their own services such as scanner, and mobile devices that became the bridge that connects between physical data that can be implemented into apparel stocks in shelves and fashion brand's digital system.

Networking tech giants like Cisco bring secure connectivity and reliable networks to handle large scale IoT deployments, while IBM leverages its strengths in AI and hybrid cloud to turn sensor data into insights. Their strategies show how competitive landscapes are shifting from simply connecting

devices to delivering meaningful analytics and business value.

Then, we have RetailNext that expertise in customer analytics. Their platforms integrate sensors, cameras, and visualization tools to help their clients, mostly fashion retailers, to understand the shopper's behavior and optimize store layouts. Overall, it is clearly visible that the market is

moving toward end to end solutions. Combining all IoT segments seamlessly as one system. Fashion retailers seek a system that able to provide all in one solution for their problem that can deliver product availability, real time demand insights and personalized shopping experience.

3. PROPOSED SOLUTION AND TECHNICAL ARCHITECTURE OVERVIEW

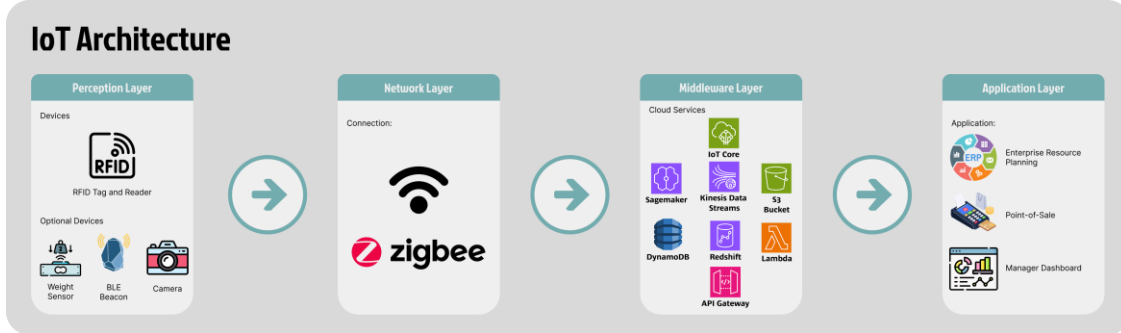


Figure 1: Four-layer IoT Architecture for RFID-enabled Demand Sensing

Based on the introduction earlier, the inefficiencies of a barcode-based fashion store, i.e. manual scanning, visibility constraints and lack of the customers' behavioral insight, is starting to get outdated and is bound for a change in the fundamentals of store operations. We propose a solution that's based on an RFID-enabled demand sensing system incorporating IoT. With the RFID tags already embedded in the Stock Keeping Unit (SKU) level, it's then seamlessly integrated into an Enterprise Resource Planning (ERP) and Point-of-Sale (POS) platforms, our solution enables continuous visibility across apparel inventory flows, supports predictive restocking using our demand sensing engine while also facilitating frictionless checkout experiences. We believe this solution is going to be relevant because our approach aligns with industry forecasts in which it said IoT in the retail sector is projected to grow from USD 66.4 billion in 2024 to USD 488.5 billion by 2033. This reflects the clear shift from hardware-focused deployments to one driven by data and is based on the cloud.

At its foundation, our system had a four-layer IoT model that we adapted specifically to fashion retail environments. The perception layer captures product and customer behavior through RFID tags, shelf-mounted readers, fitting room sensors, and optional devices such as Bluetooth beacons, cameras or weight detectors. These components together create a dense network that continuously monitors store activity in real-time. It's supplemented with a network layer by transmitting

sensor data via wireless means such as Wi-Fi and Zigbee, reasons being that the two protocols have the necessary power consumption with Wi-Fi justifying its high power usage with its high bandwidth, the required range that protocols such as Bluetooth or LoraWAN can't match, the data rate for real-time updates, the encryption protocols ensuring high security and reliability, while being cost-effective to implement.

The middleware layer hosts the data analysis part of our solution where sensor inputs are converted into states from which it's extracted into actionable insights, for example POS transaction data, RFID movement records and contextual information (e.g., seasonal events) are fused into a common pipeline.

To validate demand signals and early detection of anomalies, a data monitoring dashboard is created using AWS IoT Core, Kinesis and Grafana. This dashboard enables real-time visualization and alerts for sensor activity, the device health and inventory movement. Fuzzy logic is additionally used to handle interpretation of ambiguous insight i.e. when a product is handled by the customer but is not purchased.

Beyond edge processing, data streams are uploaded to a cloud layer built on AWS using services such as AWS IoT Core to handle device connectivity and message routing, while Amazon Kinesis Data Streams process high-volume RFID data in real-time. For long-term and offsite storage, AWS provides a service called S3 Buckets while simultaneously using DynamoDB and Redshift to

support structured queries and trend analysis. The rest are then processed by machine learning that extends the capability of demand forecasting by introducing SKU-level prediction models.

Individual items in a typical fashion store are identified using their SKU code to differentiate product variants such as their size, color or style. By leveraging this data, our system can analyze current stock availability, identify the best-selling collections and capture demand trends. The proposed machine learning approach applies five regression models i.e. Linear, Polynomial Regression, Decision Tree Regression, Support Vector Regression and Random Forest Regression, each model is tested individually per SKU.

Increasing the accuracy made by our demand sensing engine, we construct hybrid regressors by combining the two best-performing models for each SKU thus enabling adaptation to the highly varied nature of fashion demand. The inputs must include prior demand history, trend and seasonality analysis, product pricing, competitor pricing and commodity prices. This method ensures more reliable forecasts than traditional methods employed such as exponential smoothing that often fail in modern, nonlinear fashion sector. By embedding this model, the system supports better inventory management, purchasing decisions, production scheduling and resource allocation across the short-term and long-term.

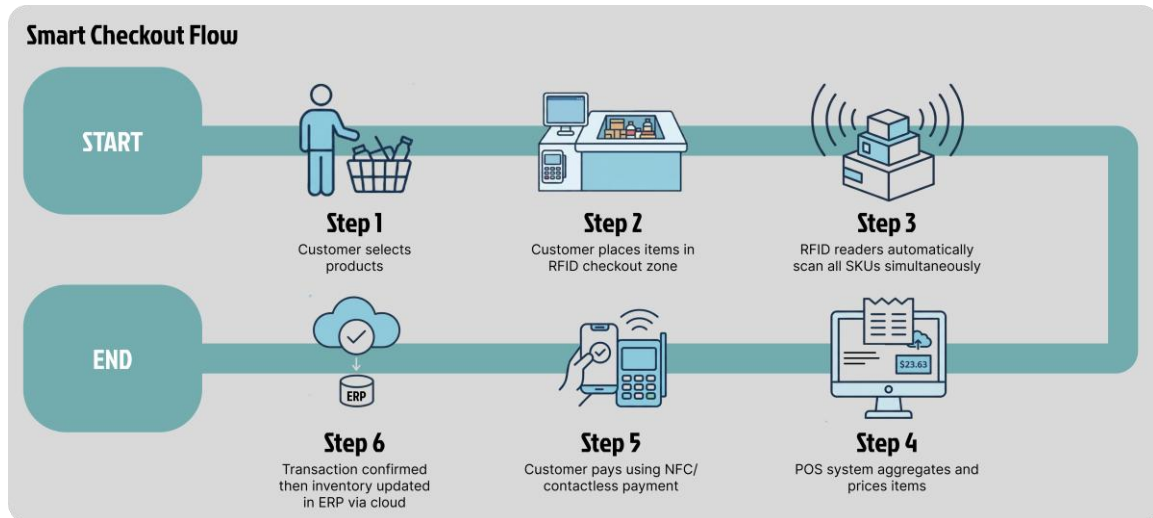


Figure 2: RFID-powered Smart Checkout Flow

One of the features that we think should be unique is its smart checkout where unlike barcodes which require direct line-of-sight and must be manually scanned, RFID readers can read multiple items at the same time. The flow for this interaction is the customers place their desired items in the checkout zone where RFID automatically scans them, and the POS generates an electronic bill to be used on contactless or NFC payments. The benefit of this feature is reducing waiting times and a unique interpretation of previous innovative models of cashierless retails such as Amazon Go, though at lower infrastructure cost since we use RFID over camera-heavy vision systems.

Complementing the checkout automation is our own demand sensing engine which can monitor SKU-level movements that include but are not limited to, products touched, tried on or purchased. These series of actions reveal to us high-frequency purchases suitable for bulk stocking, seasonal collections that experience sudden surge demand and anomalies such as sudden stockouts. Forecasts generated by this engine are streamed into an S3 bucket for persistent storage and are then distributed via AWS Lambda APIs to ERP platforms such as SAP, thus ensuring predictive insights are securely archived and accessible quickly across the network.

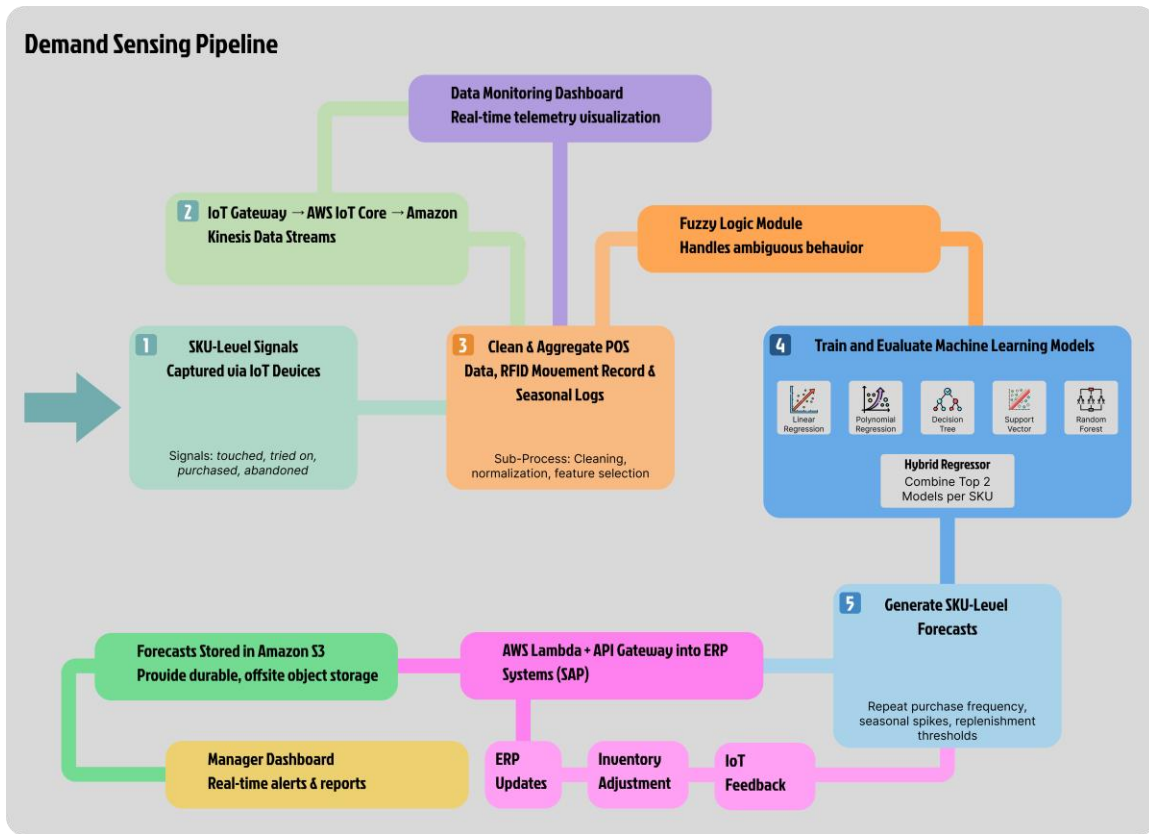


Figure 3: Demand Sensing Engine Pipeline, from SKU Activity to ERP Forecasts

The demand sensing model is strengthened by the described above machine learning models which predict repeat purchase frequencies, seasonal spikes and replenishment thresholds with improved accuracy. By relying on hybrid regressors tuned for SKU variability, the system reduces forecast error, minimizes overstocking and understocking risks and optimizes alignment between inventory and real-world demand.

By doing this implementation, it can benefit the fashion retailers, the customers and the supplier. For fashion retailers, the system provides predictive demand planning, preventing stockout and streamlined ERP operations. Customers enjoy reduced checkout friction, faster waiting times, greater product availability and personalized offers that reflect real-time shopping patterns. The suppliers gain from this have improved forecast accuracy, lower operational costs and reduced waste. With the unification of RFID, IoT and machine learning within a cloud-integrated framework, our solution addresses the shortcomings of conventional fashion retail while establishing the foundation for a more responsive, adaptive and data-driven fashion ecosystem.

4. STATE-OF-THE-ART EXAMPLES

The ideas discussed here are not merely theoretical; they have already been implemented by

major fashion retailers worldwide. These companies are actively adopting IoT and machine learning to shift from traditional operations to more intelligent, data-driven approaches.

a. UNIQLO

UNIQLO has implemented RFID technology in its stores for automatic payment processes that allow customers to pay independently. Thanks to this automated system, UNIQLO has managed to reduce transaction times by around 50%. However, the greatness of this system does not stop there. Product movement data, for example, items that customers often try on but ultimately do not purchase, are also analyzed in depth. This valuable information enables UNIQLO to predict demand in detail, adjust inventory at each store, and ultimately minimize issues of overstocking or understocking. As a result, product availability on the shelves truly reflects consumer demand.

b. Zara

Zara's fashion supply chain is renowned for its speed and agility, largely due to its use of IoT technology. The success of Zara's supply chain is driven by using real-time data from IoT technology. Zara is equipped with RFID tags for tracking every single item from the warehouse to the store shelf. By linking RFID data with sales data from cash

registers, Zara's system can quickly identify rising trends. Consequently, the firm can immediately change production schedules, reorder stock, and distribute popular products to stores with the highest demand. This almost immediate demand-sensing method effectively avoids the need for large sales while ensuring that collections remain fresh and up-to-date with current trends.

c. Walmart

Walmart has leveraged RFID technology specifically to revamp its apparel and general merchandise departments. By requiring RFID tagging for its apparel suppliers, Walmart ensures that store employees can verify inventory accuracy for thousands of items in minutes, not hours. This technology addresses a critical challenge in fashion retail: ensuring that specific sizes and colors are available on the shelf, rather than sitting in a warehouse. By improving On-Shelf Availability (OSA), Walmart has significantly increased its

apparel sales and improved the efficiency of its omnichannel fulfillment (integration of physical stores, apps, websites, and social media), ensuring that customers can find the exact styles they are looking for.

Like other major players in the industry, we use IoT sensors in RFID to track inventory. However, we utilize this real-time data more intelligently. Our machine learning (ML) models do not use a one-size-fits-all approach but rather create and combine the best models for each SKU individually. This enables much more accurate demand predictions for each product variant, such as color, size, etc., overcoming the limitations of traditional models. The result is a system that not only optimizes stock like UNIQLO, Zara, and Walmart's apparel division, but also simplifies operations with RFID-based automated checkout, an innovative leap that makes advanced fashion retail experiences more affordable and implementable.

Table 1: Comparative Table: State-of-The-Art vs Proposed Solution

Metrics	UNIQLO	Zara	Walmart	Proposed Solution
Main Technology	RFID	RFID & IoT	RFID	IoT (RFID, sensor, camera), Cloud AWS services, Network, and Machine Learning
Implementation Focus	Automatic payment process (self-checkout).	Supply chain that's renowned for its speed and agility due to the use of IoT technology.	Apparel inventory accuracy and on-shelf availability tracker.	Smart Checkout & Demand Sensing Engine (Demand Prediction)
Data Analytics Aspect/Method	Demand prediction analysis includes both frequently purchased and unpurchased items to minimize stock count errors.	Using RFID linked to sales data can analyze trends in products that customers frequently and rarely purchase.	Rapid inventory counting that verifies size and color availability on the sales floor versus the backroom stock.	Hybrid ML Regressors tailored to each SKU of each item, fuzzy logic to handle ambiguous data (such as products touched but not purchased), and real-time integration of point of sale (POS) data, RFID, and seasonal data for SKU-level forecasting.
Benefits & Improvements	Reduces transaction time (around 50%) and optimizes stock.	Responding quickly to trends increases productivity and product sales while reducing excess inventory.	Ensuring specific styles remain available, correcting stock inaccuracies (replenishing sizes), and boosting apparel sales through better on-shelf availability.	Reduce forecasting errors and minimize the risk of overstocking or understocking. This benefits retailers through predictive planning and stock-out prevention, customers through faster checkout and improved product availability, and suppliers through improved forecasting accuracy, reduced operational costs, and waste.

Table 1 compares three state-of-the-art (SOTA) retail solutions: UNIQLO, Zara, and Walmart, compared to the proposed solution in this study. SOTA companies primarily utilize RFID, with Zara also incorporating IoT for specific operational focuses. UNIQLO focuses on automated self-checkout, Zara on supply chain agility to respond to fast-fashion trends, and Walmart on apparel inventory accuracy and on-shelf availability. The proposed solution, on the other hand, utilizes a broader technology stack (IoT, AWS Cloud, machine learning, and networking) for two key focuses: smart checkout and a demand sensing engine (demand forecast).

The proposed solution overcomes the limitations of SOTA by shifting from simple operational tracking to sophisticated predictive analytics (forecasting). The primary problem it solves is forecasting errors, which lead to overstocking (deadstock) or understocking. Unlike SOTA, which tracks general trends, this solution uses hybrid machine learning regressors specifically designed for each SKU and each item, enabling much more accurate predictions. Additionally, this solution uses fuzzy logic to handle ambiguous data (such as products touched but not purchased) and integrates POS, RFID, and seasonal data in real time. This solution provides deeper and more comprehensive benefits for retailers (predictive planning), customers (product availability), and suppliers (waste reduction).

5. CONCLUSIONS

This study proposes a fundamental transformation in fashion retail operations by integrating a four-layer IoT architecture with RFID technology to address the visibility gaps inherent in traditional barcode systems. Central to this solution is a sophisticated demand-sensing engine that leverages machine learning to move beyond simple tracking. By evaluating five distinct regression models, including linear, polynomial, decision tree, support vector, and random forest. Then, synthesizing the top two performers into a hybrid predictor for every individual SKU, the system achieves superior forecasting accuracy tailored to specific product variants. Furthermore, the inclusion of fuzzy logic allows the system to interpret ambiguous customer behaviors, such as items tried on but abandoned, converting raw sensor data into actionable insights within a robust AWS cloud ecosystem.

The implementation of this data-driven architecture delivers comprehensive benefits across the entire fashion value chain. Fashion retailers gain predictive capabilities that align inventory with real-time context, significantly reducing deadstock risks and automating ERP updates. For customers, the integration supports a frictionless Smart Checkout experience where RFID readers scan multiple items simultaneously, drastically cutting transaction times and ensuring desired products are always available. Simultaneously, suppliers benefit from high-precision forecasts that lower operational costs and minimize waste generated by unsold inventory.

Ultimately, this solution navigates the critical challenges of the modern market, specifically the inability of conventional forecasting methods to handle nonlinear fashion demand trends and the operational inefficiencies of manual scanning. By transitioning from static hardware deployments to a dynamic, data-centric platform, this approach not only resolves immediate operational friction but also secures a competitive advantage through continuous, context-aware intelligence, establishing a responsive foundation for the future of fashion.

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